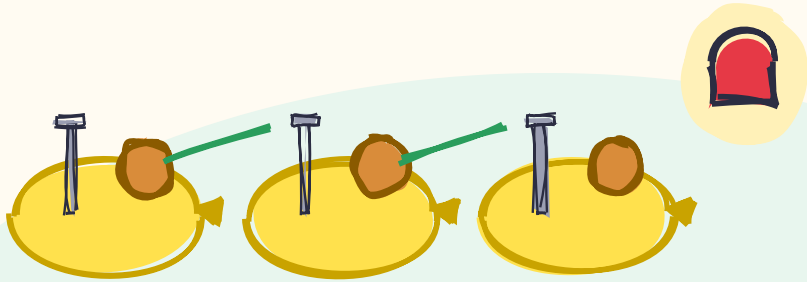
Little Explorers · Class 3

Lemon Battery LED Lamp

Can a lemon light a lamp? Let's find out!

Lemons | Nail + coin | Chain them up!



Did you know?
A potato, an apple or an orange can make electricity too. Which do you think is best?

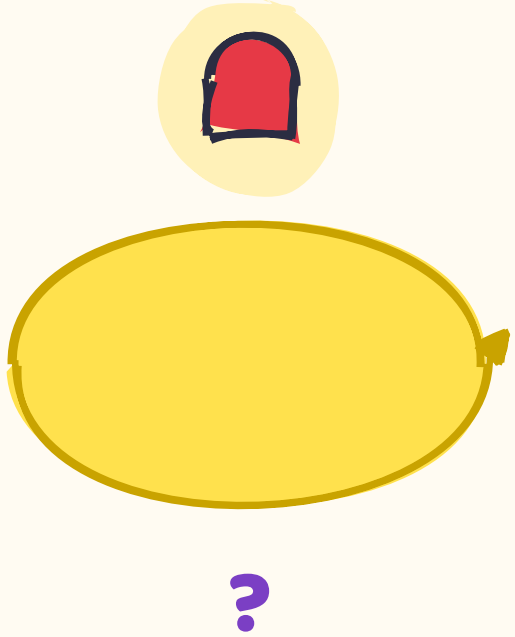
LEARN

Let's wonder...

Is there electricity in a lemon?

A sour surprise

Batteries come from shops. But what if we could make one in our classroom from **fruit**? Can a lemon really make a light glow?



No way!

Maybe a tiny bit?

Let's test it!

Today: we turn lemons into a **battery** and try to light an **LED**!

Learning objectives

Today I can…

1	**Energy:** say that a battery **stores energy** to use later	**Proof:** tell a friend one thing that uses a battery
2	**Build:** make a **lemon cell** with a nail and a coin	**Proof:** my lemon has a nail and a coin pushed in
3	**Team up:** join **more lemons** in a chain to light the LED	**Proof:** our LED glows and we counted the lemons

Objective 1 · Energy · video

Video time!



How Batteries Work for Kids

Socratica Kids · about 6 minutes · youtu.be/RWWLfUIiMAw

Watch for...

1. What does a battery **store**?
2. What are the two ends of a battery called?
3. Name **three things** that use batteries.

Scan to watch

After the video: Point to the + end and the - end of a real battery. Tell your partner what each one means.

Objective 1 · Energy

A battery is an energy lunchbox

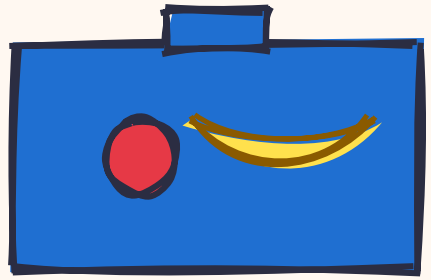
The idea

A battery **stores energy** inside it. When we join it into a circuit, the energy comes out as **electricity**.

Every battery has two ends: **+** (plus) and **-** (minus).

Safety
Never open a real battery. The inside can hurt your skin.

Example: your lunchbox



Mum packs your lunchbox in the morning. The food **stores energy** for later. At lunchtime you eat it and get energy to play! A battery saves energy for later too.

Batteries everywhere

What uses a battery?



Toys
Cars, robots and dolls that move or talk



Phones
A big flat battery hides inside



Clocks
A small battery keeps time for a year



Torches
Light in the dark, anywhere you go

? What would happen if all batteries stopped working today?

Count them
How many things with batteries are in your home?

Objective 3 · Team up

One lemon is not enough

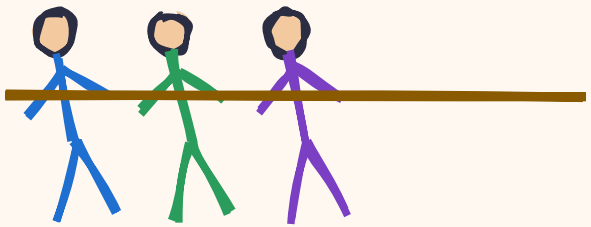
The idea

One lemon makes a **tiny** bit of electricity: not enough for an LED.

So we make a **team**! We join lemons in a chain: **coin** of one lemon to the **nail** of the next. Each lemon adds a little more push.

Did you know?
Real batteries are also made of small cells joined together!

Example: tug of war



One child cannot pull the big rope. But a **team** of children pulling **together** can win! Lemons in a chain work as a team.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Lemon lab rules
2. Make one lemon cell
3. Squeeze chain
4. Chain the lemons and light the LED
5. How many lemons do we need?
6. Fruit battery challenges

Stay safe

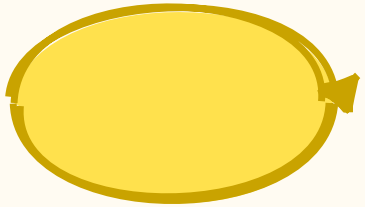
Lemon lab rules

Do not eat the lemons after: they have metal in them.

Careful with nails. Push slowly, point away from fingers.

Wash hands after the lab. Lemon juice stings in eyes!

Recycle put lemons in the compost bin at the end.



Remember

Fruit batteries are safe. Wall plugs are NOT. Never play with sockets!

MAKE

Objective 2 · Build step 1 of 2

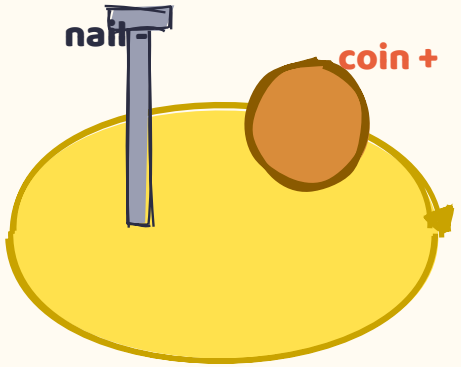
Make one lemon cell

a Roll the lemon on the table to make it **juicy** inside.

b Push a **nail** into one side. A grown-up can help.

c Push a **coin** into the other side. Do not let them touch!

d Now you have a **cell**: the coin is **+** and the nail is **-**.



Check: The nail and the coin are both inside the lemon, not touching each other.

Nails are sharp. Push slowly and point them away from fingers.

Class game

Squeeze chain

How to play

1. Stand in a line and hold hands.
2. The first child **squeezes** the next hand.
3. Pass the squeeze down the line as fast as you can.
4. The last child shouts **LIGHT!**
5. If one child lets go, the squeeze stops!

Then discuss

What happens if one person **lets go**? Is that like a broken wire?



Big idea

Electricity needs a full, unbroken chain to reach the light.

Objective 3 · Build step 2 of 2

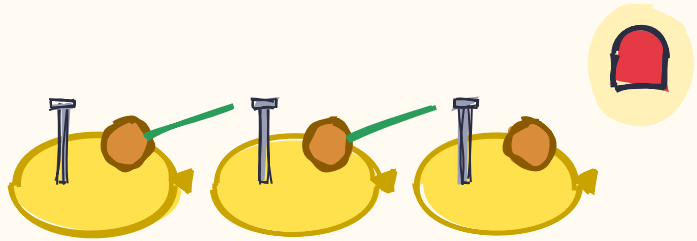
Chain the lemons and light the LED

a Clip the **coin** of lemon 1 to the **nail** of lemon 2. Do the same for lemon 3 and 4.

b The **long leg** of the LED goes to the **coin** at one end of the chain.

c The **short leg** goes to the **nail** at the other end.

d Turn off the room lights. Cup your hands around the LED. Is it glowing?



Check: The LED glows a little (red works best). Look closely in the dark!

No glow? Turn the LED round, check every clip, or add another lemon.

Experiment!

How many lemons do we need?

Lemons	I think it glows…	What we saw
1 lemon	yes / no?	_____
2 lemons	yes / no?	_____
3 lemons	yes / no?	_____
4 lemons	yes / no?	_____

Talk about it

Did more lemons make a **brighter** light? Why do you think so?

Make it fair

Use the same LED and the same clips every time.

Level up!

Fruit battery challenges

Choose your challenge

- **Fruit race:** try a potato, an apple or an orange. Which glows best?
- **Mega chain:** join with another team. Does 8 lemons glow brighter?
- **Colour test:** try a blue LED. Does it need more lemons?
- **Squeeze test:** does a squashed, juicy lemon work better?

Think like an engineer

Where does the lemon's energy really come from?

Boss level

Light the LED with the fewest lemons possible!

Objective check

Can I really do it? Prove it.

01	Tell your partner one thing a battery does.	○ ○ ○
02	Show your lemon cell: where is the nail? Where is the coin?	○ ○ ○
03	How many lemons did your team need to light the LED?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Make a class bar chart: number of lemons each team needed.

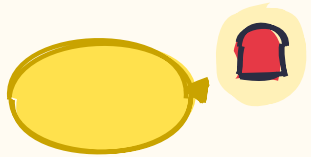
Show what you know

What did we learn?

- [x] A battery stores energy for later
- [x] A lemon, a nail and a coin make a cell
- [x] One cell is weak: join cells in a chain
- [x] The LED's long leg goes to the coin (+)

At home

Ask a grown-up to show you a battery. Find the + and − ends.



Exit ticket: why did we need more than one lemon?

Next class: build a Traffic Light!